Electrolysis of Aqueous Sodium Chloride & Dilute Sulfuric Acid

Exam code: 0620 & 0971

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Electrolysis of aqueous sodium chloride & dilute sulfuric acid

Electrolysis of aqueous sodium chloride

- **Brine** is a concentrated solution of aqueous sodium chloride
- It can be electrolysed using inert electrodes made from platinum or carbon / graphite
- The ions in brine are:
  - Na^+ and Cl^- ions from the brine / aqueous sodium chloride
  - H^+ and OH^- ions from the water
- When electrolysed, it produces bubbles of gas at both electrodes
  - The gases chlorine and hydrogen are produced
  - Sodium hydroxide solution is the product remaining in the electrolysis chamber / container
- These substances all have important industrial uses:
  - Chlorine is used to make bleach
  - Hydrogen is used to make margarine
  - Sodium hydroxide is used to make soap and detergents

Product at the negative electrode:

- The H^+ ions are discharged at the **cathode** as they are less reactive than sodium ions
- The H^+ ions gain electrons to form **hydrogen** gas

Product at the positive electrode:

- The Cl^- ions are discharged at the **anode**
- They lose electrons and **chlorine gas forms**
- The Na^+ and OH^- ions remain behind and form the NaOH solution

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

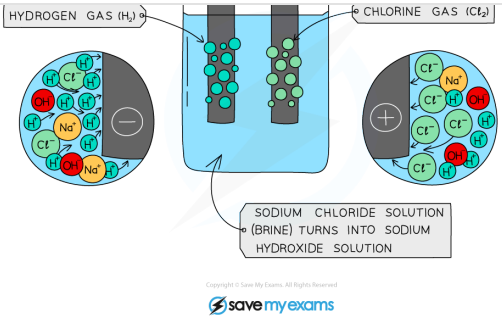
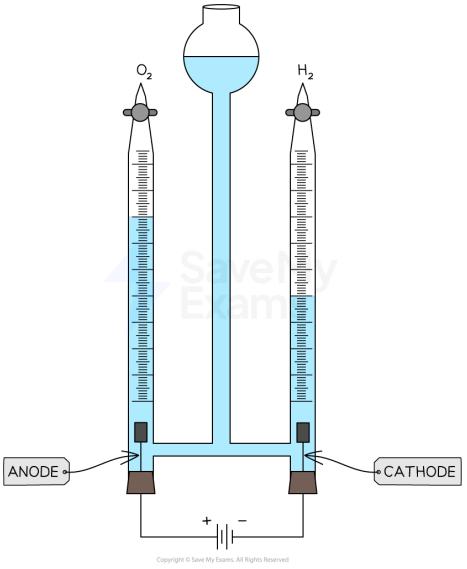


Diagram showing the products of the electrolysis of aqueous sodium chloride

Electrolysis of dilute sulfuric acid

- Dilute sulfuric acid can be electrolysed using inert electrodes made from platinum or carbon / graphite
- The ions in dilute sulfuric acid are:
  - H^+ and SO_4^{2-} ions from the sufuric acid
  - H^+ and OH^- ions from the water

- When electrolysed, it produces bubbles of gas at both electrodes
  - The gases oxygen and hydrogen are produced



Electrolysing dilute sulfuric acid in a Hoffman voltameter shows that twice as much hydrogen is produced, compared to oxygen

Product at the negative electrode:

- H^+ ions are attracted to the **cathode**
- The H^+ ions gain electrons to form **hydrogen** gas

Product at the positive electrode:

- OH^- ions are attracted to the **anode**
- They lose electrons and form **oxygen gas** and water

Examiner Tips and Tricks

- When a gas is produced during electrolysis, you should be able to give the appropriate gas test:
  - Oxygen - if a glowing splint is dipped into a sample of the gas, then the splint will relight
  - Hydrogen - if a lit splint is dipped into a sample of the gas, then a squeaky pop will be heard
  - Chlorine - if damp litmus paper is dipped into a sample of the gas, it will turn red and then bleach to a white colour

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